Do Now · retrieval

Logic circuits, adders and flip-flops

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why can a group of four 1s remove two variables?

Q2

Simplify ¬(A ∧ B).

Q3

Calculate 1+1 in binary.

Q4

What is (1 XOR 0) AND 1?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Find the Sum bit — XOR is 1 when A and B are different; A=1, B=1 gives Sum=0.

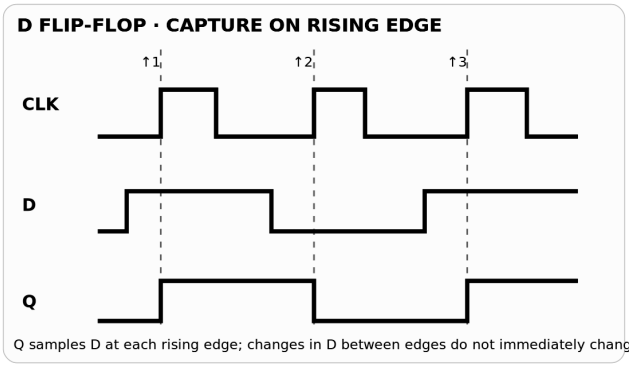
Core idea 2

Find the Carry bit — AND is 1 only when both inputs are 1; here Carry=1.

Core idea 3

Extend to full adder and memory — A full adder also includes CarryIn. Unlike combinational logic, a D flip-flop stores a bit until the triggering clock edge.

Key visual · complete concept



New Learning · applying the ideas

Vocabulary · use the correct technical terms

half adder — Combinational circuit that adds two one-bit inputs.

full adder — Adds two bits and a carry-in bit.

Sum — Current-column result bit.

CarryOut — Bit passed to the next higher column.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

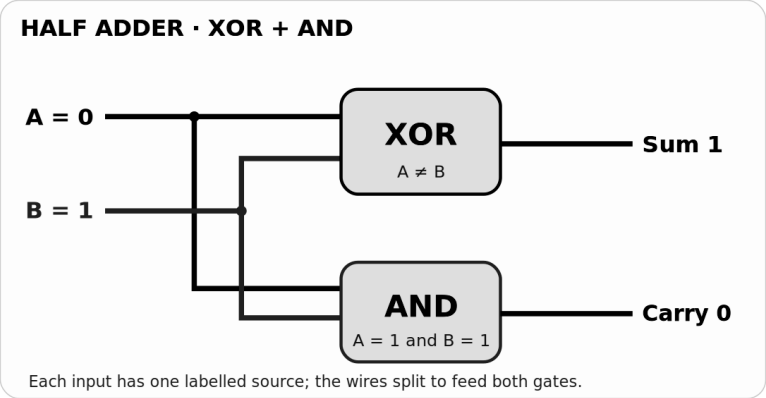
Problem

A half adder receives A=0, B=1. Calculate Sum and Carry.

Follow the worked method

1. Predict and explain — XOR(0,1)=1, giving Sum=1.
2. Check and extend — AND(0,1)=0, giving Carry=0.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict and explain
The total is decimal 2, binary 10.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

A half adder receives A=1, B=0. Give Sum and Carry.

Question 2

For a full adder A=1, B=1 and CarryIn=0, calculate Sum and CarryOut.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A rising-edge D flip-flop has Q=0 and D=1. D changes to 0 before the next rising clock edge. What value is stored after that edge, and why?

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A D-type flip-flop always changes Q immediately when D changes.

Use key vocabulary from memory

1. Explain half adder in your own words.

2. Explain full adder in your own words.

3. Explain Sum in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.